

IKBKE fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes IKBKE gene fusions from the msk_impact_50k_2026 study, covering 12 fusion events. 3/12 fusions (25.0%) were in-frame. 4/12 fusions (33.3%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.0242424$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for IKBKE.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 12 analyzed fusion events, identifying 11 distinct fusion partner genes. The most frequent partner was NAV1, observed in 2 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 3/3 (100.0%) of in-frame fusions retained the domain, $p=0.0242424$ (statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.019802.

Domain disruption

Domain-disruption analysis was skipped: IKBKE has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 11 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 359 aa (exon 10; permutation-corrected $p=0.39604$, not statistically significant at $\alpha=0.05$). This is 115 aa from the nearest configured domain boundary at 244 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=4$) and "not_retained" ($n=7$). For Frame_status, retained: 3/4 (75.0%, 95% CI 30.1%-95.4%); not_retained: 0/2 (0.0%, 95% CI 0.0%-65.8%). Welch's t-test comparing Tumor_variant_count between groups gave means of 19.75 vs 19.4286, $p=0.973661$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for IKBKE in this cohort; expression-association analysis was skipped.

Joint partner

Joint partner reported: IKBKE has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: IKBKE has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (10 row(s) in window_scan, 5 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RASAL2	1	0.08333333333333333	0.17032913781186615	None	1.0	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.3150159387473804	1
NAV1	2	0.16666666666666666	0.17032913781186615	None	0.7275541795665634	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.295182347416111	2
NFASC	1	0.08333333333333333	0.17032913781186615	None	0.8885448916408669	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.29411810593004295	3
TCEA3	1	0.08333333333333333	0.17032913781186615	None	0.5263157894736843	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2262001492736962	4
RASSF5	1	0.08333333333333333	0.17032913781186615	None	0.44272445820433437	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.21052677466069308	5
SRGAP2	1	0.08333333333333333	0.17032913781186615	None	0.3405572755417957	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.19137042791146705	6
RBBP5	1	0.08333333333333333	0.17032913781186615	None	0.26006191950464397	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.17627754865450113	7
SMYD2	1	0.08333333333333333	0.17032913781186615	None	None	0.34430466544937777	confidence_certainty, domain_retention, recurrence	0.1569426938429297	8
THBS3	1	0.08333333333333333	0.17032913781186615	None	0.12383900928792568	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.15073575298886646	9
IL20	1	0.08333333333333333	0.17032913781186615	None	0.0	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.12751593874738038	10
RABGAP1L	1	0.08333333333333333	0.17032913781186615	None	0.0	0.34430466544937777	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.12751593874738038	11

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
76	0	4	1	6	0.0	1.0	-0.0
120	0	4	2	5	0.0	0.4909090909090909	0.30899892533525647
206	0	4	3	4	0.0	0.23636363636363636	0.626419337187407
359	0	4	4	3	0.0	0.19393939393939394	0.7123339658940003
395	1	3	4	3	0.25	0.5454545454545454	0.26324143477458145
447	3	1	4	3	2.25	1.0	-0.0
539	3	1	5	2	1.2	1.0	-0.0
572	3	1	6	1	0.5	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

3	1
0	7

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	11	4	3	4	0.13191489361702127

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
3	6	12	0.5

frequency tables

Partner_gene_counts

Partner_gene	Event_count
IL20	1
NAV1	2

Partner_gene	Event_count
NFASC	1
RABGAP1L	1
RASAL2	1
RASSF5	1
RBBP5	1
SMYD2	1
SRGAP2	1
TCEA3	1
THBS3	1

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
247	447	200	4	7	3	1	18.0	0.08787878787878788	1.0561159419789314	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16
339	539	200	5	6	3	1	7.5	0.24242424242424238	0.615423952885944	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25
347	447	100	4	7	3	1	18.0	0.08787878787878788	1.0561159419789314	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16
359	459	100	4	7	3	1	18.0	0.08787878787878788	1.0561159419789314	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16
359	559	200	5	6	3	1	7.5	0.24242424242424238	0.615423952885944	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25
372	572	200	5	6	3	1	7.5	0.24242424242424238	0.615423952885944	EVT-P-0001902-T01-IM3-23, EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25
395	595	200	5	6	3	1	7.5	0.24242424242424238	0.615423952885944	EVT-P-0001902-T01-IM3-23, EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
447	647	200	4	7	2	2	2.5	0.5757575757575757	0.23976033892505857	EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25
482	682	200	4	7	1	3	0.4444444444444444	1.0	-0.0	EVT-P-0022004-T02-IM6-27, EVT-P-0027193-T03-IM6-14, EVT-P-0053124-T01-IM6-17, EVT-P-0067348-T01-IM7-25
539	739	200	4	7	1	3	0.4444444444444444	1.0	-0.0	EVT-P-0022004-T02-IM6-27, EVT-P-0027193-T03-IM6-14, EVT-P-0053124-T01-IM6-17, EVT-P-0067348-T01-IM7-25

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
347	447	100	4	7	3	1	18.0	0.08787878787878788	1.0561159419789314	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16
339	539	200	5	6	3	1	7.5	0.24242424242424238	0.615423952885944	EVT-P-0001902-T01-IM3-23, EVT-P-0019364-T01-IM6-24, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25
372	572	200	5	6	3	1	7.5	0.24242424242424238	0.615423952885944	EVT-P-0001902-T01-IM3-23, EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25
447	647	200	4	7	2	2	2.5	0.5757575757575757	0.23976033892505857	EVT-P-0022004-T02-IM6-27, EVT-P-0051760-T01-IM6-15, EVT-P-0051760-T02-IM6-16, EVT-P-0067348-T01-IM7-25
482	682	200	4	7	1	3	0.4444444444444444	1.0	-0.0	EVT-P-0022004-T02-IM6-27, EVT-P-0027193-T03-IM6-14, EVT-P-0053124-T01-IM6-17, EVT-P-0067348-T01-IM7-25

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 7193-T03-IM6-14	P-002 7193-T03-IM6		Prostate Cancer	PRA D	IKBKE-IL20	IL20	unknown	three_prime	206667107	21	682	True	682	True	exact match	lost	0.0	False		Protein kinase domain		IL20 (NM_018724) - IKBKE (NM_014002) Rearrangement : c.398:1L20_c.2046-146:1KBKE Protein Fusion: mid-exon {IL20:1KBKE}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005-1760-T01-IM6-15	P-005-1760-T01-IM6		Melanoma	MEL	IKBKE-NAV1	NAV1	in-frame	five_prime	206653556	12	447	True	446	True	hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - NAV1 (NM_020443) rearrangement: c.1340+100:1 KBKE_c.126+27872: NAV1 dup Protein Fusion: in frame {IKBKE:NAV1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005-1760-T02-IM6-16	P-005-1760-T02-IM6		Melanoma	SKCM	IKBKE-NAV1	NAV1	in-frame	five_prime	206653558	12	447	True	446	True	hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - NAV1 (NM_020443) rearrangement: c.1340+102:1 KBKE_c.1226+27873: NAV1 dup Protein Fusion: in frame {IKBKE:NAV1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 3124-T01-IM6-17	P-005 3124-T01-IM6		Hepatobiliary Cancer	GBC	IKBKE-RA BGA P1L	RAB GAP1 L	out-of-frame	five_prime	206666746	20	682	True	681	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBKE (NM_014002) - RAB GAP1 L (NM_014857) rearrangement: c.2045+35:1 KBKE_c.1710+64980: RAB GAP1 Ldup Protein Fusion: out of frame {IKBKE:RA BGA P1L}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005-2992-T01-IM6-18	P-005-2992-T01-IM6		Melanoma	SKCM	IKBKE-RB-BP5	RBBP5	out-of-frame	five_prime	206648412	5	120	True	119	True	hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.4723404255319149	True			Protein kinase domain	IKBKE (NM_014002) - RBBP5 (NM_005057) rearrangement: c.358+75:IKBKE_c.218+215:RBBP5in v Protein Fusion: out of frame {IKBKE:RB-BP5}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 8226-T01-I M6-21	P-005 8226-T01-I M6		Breast Cancer	IDC	IKBK E-TH BS3	THBS3	unknown	five_prime	206648064	4	76	True	76	True	exact match	disrupted	0.2851063829787234	True			Protein kinase domain	IKBKE (NM_014002) - THBS3 (NM_007112) rearrangement: c.229-144:IKBKE_c.2579:THBS3inv Protein Fusion: mid-exon {IKBKE:THBS3}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-000 1902-T01-I M3-23	P-000 1902-T01-I M3		Breast Cancer	ILC	NFASC-IKBBKE	NFASC	in-frame	five_prime	206652479	10	395	True	394	True	hybrid junction condition: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	retained	1.0	False	Protein kinase domain			IKBBE (NM_014002) - NFASC (NM_001005388) Duplication : c.1183+3_c.2782+2783dup Protein fusion : in frame (IKBBE-NFASC)	NA

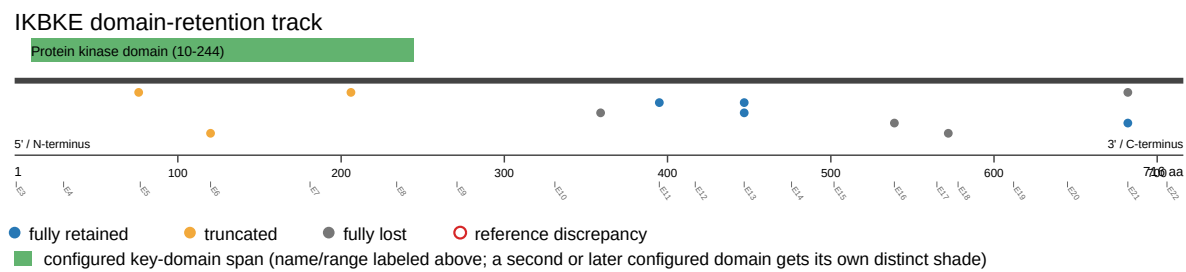
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001-9364-T01-IM6-24	P-001-9364-T01-IM6		Breast Cancer	IDC	RASAL2-IKBKE	RASAL2	unknown	three_prime	206652369	10	359	False	359	True	exact match	lost	0.0	False		Protein kinase domain		RASAL2 (NM_170692) - IKBKE (NM_014002) Rearrangement : c.1628-181:RASAL2_c.1076:IKBKEdel Protein Fusion: mid-exon {RASAL2:IKBKE}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0067348-T01-IM7-25	P-0067348-T01-IM7		Ovarian Cancer	HGSO	RASSF5-IKBKE	RASSF5	out-of-frame	three_prime	206661210	16	539	True	539	True	exact match	lost	0.0	False		Protein kinase domain		RASSF5 (NM_182663) - IKBKE (NM_014002) rearrangement: c. 579+1935: RASSF5_c. 1617-41:IKBKEdup Protein Fusion: out of frame {RAS SF5:IKBKE}	NA

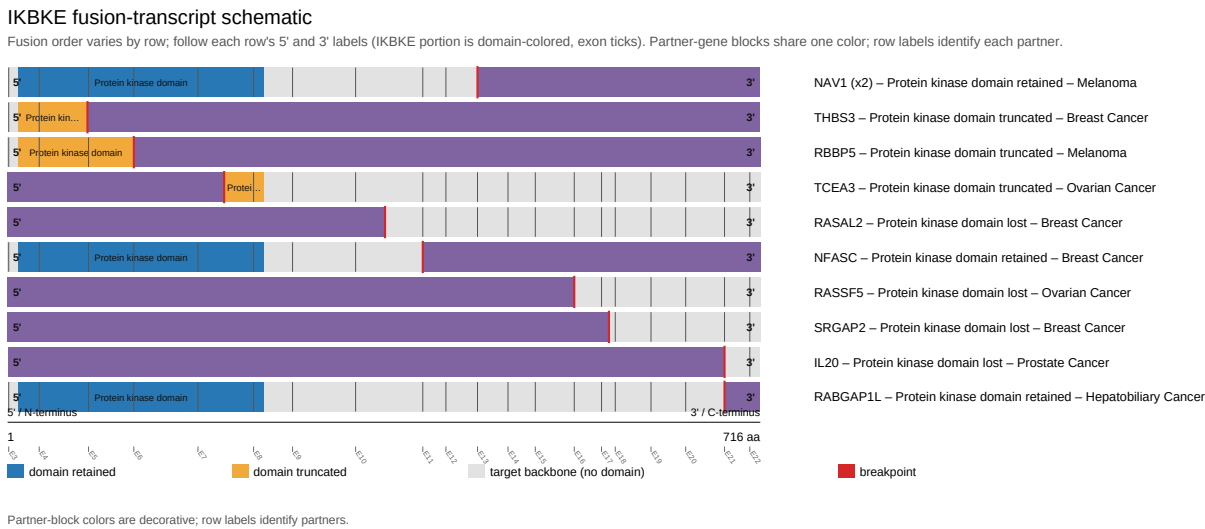
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-002 2004-T02-I M6-27	P-002 2004-T02-I M6		Breast Cancer	IDC	SRG AP2-I KBKE	SRG AP2	unknown	three_prime	206664172	17	572	False	572	True	exact match	lost	0.0	False		Protein kinase domain		SRG AP2 (NM_015326) - IKBKE (NM_014002) rearrangement: c.2085:SRGAP2_c.1714:IKBK Edell Protein Fusion: miRd-exon {SRGAP2:IKBKE}	NA
EVT-P-005 4307-T01-I M6-28	P-005 4307-T01-I M6		Ovarian Cancer	HGS OC	TCEA3-IKBE	TCEA3	unknown	three_prime	206650096	7	206	False	206	True	exact match	disrupted	0.16595744680851063	True			Protein kinase domain	TCEA3 (NM_003196) - IKBKE (NM_014002) rearrangement: c.69+346:TC EA3_c.616:IKBK Einv Protein Fusion: miRd-exon {TC EA3:IKBKE}	NA

Figures

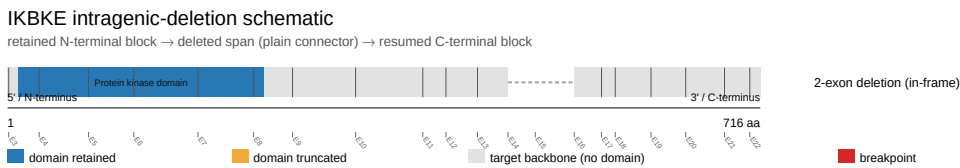
domain retention outliers



fusion schematic



intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

